Peak-Hour Boundary Detection

When does rush hour *really* start? An online concept-drift study of the Salvador transit network

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Concept Drifters
Aryan Eugene Hans Necmi Saurabh Van

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University of Waikato

The problem: schedules are static, demand is not

Transit runs on fixed peak windows, e.g. 06:00–09:00, 17:00–20:00.

But the **real** demand peak at a stop can start earlier or later, differ by stop, and shift with weekday / weekend / weather.

When a bus arrives *after* the peak, the platform queue grows:

$$Q(t) = \int_{t_0}^{t} \big(\rho(\tau) - \kappa(\tau)\big)\, d\tau$$

with arrival rate $\rho(t)$ (demand) and scheduled capacity $\kappa(t)$ (supply).

If κ lags ρ, then $\frac{d}{dt} Q(t) > 0$: overcrowding and downstream congestion. Static GTFS tables cannot see this in real time.

Our questions, answered as a data stream

1. Where is each stop's **real** peak (not the official window)?
2. Does that temporal boundary exhibit **concept drift** over our time-frame?
3. Does the **timetable** actually match the demand peak?

Hard constraint
Models must be **online / streaming** (MOA / CapyMOA), no batch. Everything is interleaved *test-then-train* (prequential), single-pass.

Data: the SUNT 10-stop, 5-minute subset

`labiaufba/SSA_StopBusTimeSeries_5` (Salvador, Brazil).

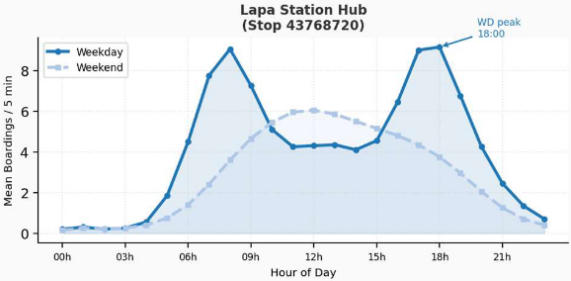
- **92 days**, March–May 2024 (66 weekday, 26 weekend); 5-min aggregation.
- Three parallel series per stop: *boarding*, *landing*, *loader* (occupancy).
- Streamed in $\mathcal{O}(1)$ chunks, no CSV fully held in RAM.
- 16.51% missing 5-min slots imputed with a leakage-free EWMA ($\alpha = 0.05$, real observations only; imputed rows flagged).

Combined with Open-Meteo weather + Bahia public holidays as external covariates – compiled to `features_stream.csv`.

Method: four online components

1. **Prequential demand forecasting**: CapyMOA Adaptive Random Forest / FIMTDD / Passive-Aggressive vs. naive baselines (windowed RMSE / MAE).
2. **Adaptive peak detection**: online EWMA baseline; an interval is "peak" (in the sense of rush-hour) when the rate exceeds $\tau_t = \mu_{t-1} + k\,\sigma_{t-1} + \delta$, sustained ≥ 20 min $\Rightarrow$ data-driven peak start.
3. **Drift tracking**: CapyMOA ADWIN on the boundary series and on the model's error.
4. **Schedule alignment**: from GTFS `stop_times.txt`, the scheduled peak vs. the demand peak (the actionable gap).

EDA: one window does not fit all stops



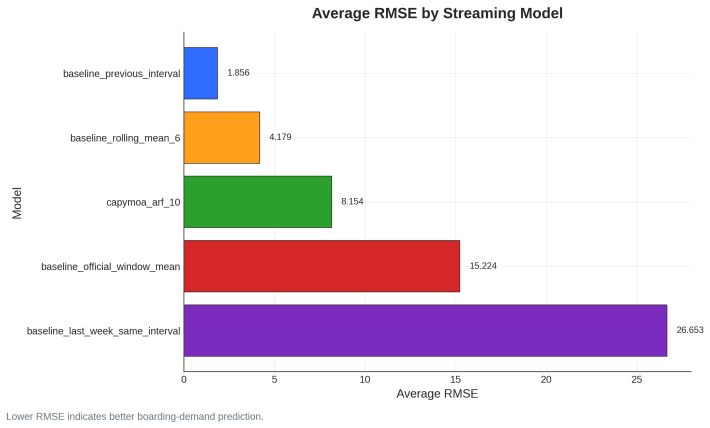
Lapa Station: weekday twin peaks

Three archetypal stops:

- **UFBA** commuter: weekday peak ∼9.1 @ 08:00, 6.5× weekday/weekend.
- **Lapa** hub: twin peaks 07:00 & 17–18:00 (≈10.2, the max).
- **Itapua** beach: weekend midday plateau, busiest at 12:00, not 08:00.

Fixed 06:00–09:00 / 17:00–20:00 windows misclassify ≥7 of 10 stops.

Result 1: demand is predictable online



Lower RMSE indicates better boarding-demand prediction.

- 5-min demand is highly autocorrelated $\Rightarrow$ the **previous-interval** baseline is very strong. Suggesting that boarding demand is *generally* **Markovian**.
- Best CapyMOA model: **Passive-Aggressive**, which *beats* the baseline at the busy Lapa hub (MAE 6.65 vs 7.22).
- ARF gives a valid comparison; SGD / FIMTDD unstable without tuning (reported honestly).

  Evaluated strictly test-then-train (no look-ahead).

Result 2: is there real concept drift?

- ADWIN on the boundary series (Framing 1): 0 temporal drift (peak demand time stable).
- ADWIN on the model's per-day RMSE (Framing 2): 0 drift (model's ϵ-baseline stable).
- Per-instance ADWIN fires almost daily, but that is strong daily *seasonality*, not model decay (a methodological caveat, stated openly).

Honest Findings
Since the 3-month window exhibits **zero temporal drift or concept drift**, our contribution focuses on the **structural operational divergence**: measuring exactly where the static GTFS schedule fails the true, stop-specific peaks.

Result 3 (headline): the timetable is ahead of demand

Morning (AM) window, the trustworthy comparison (sharp ramp on both sides):

Stop	Demand peak	Scheduled peak	Gap
Lapa Station	06:09	05:50	+19 min
Salvador Bus Station	06:20	05:45	+35 min
Ferry terminal	06:46	06:00	+46 min
Castro Alves Theater	07:21	06:30	+51 min
Lacerda Elevator	07:27	06:25	+63 min

Demand *always* arrives **after** the scheduled peak: the timetable front-loads morning service by ∼20–60 min. (PM is a long plateau, so PM gaps are reported with caution.)

If nothing is drifting, why is the network failing?

The Paradox: ADWIN proved there are no temporal or concept drifts. Commuter routines are predictable overall.

The Cause (Misalignment):

- **The Global Assumption:** GTFS provides a supply function $\kappa(t)$ derived from batch methods. It is blind to real-time node variance.
- **Empirical Observations:** The true demand stream $\rho(t)$ is partitioned by spatial regimes (Lapa peaks early, Itapua peaks midday, weekend volumes crash).
- **The Expected Reaction:** Because the static $\kappa(t)$ needs to be ahead of $\rho(t)$ but utilizes a generic $\sim$06:00 network-wide schedule rush, it excessively over-estimates the time of local demand rush ($\kappa(t) \gg \rho(t)$), forcing the $+19$ to $+63$ minute gaps.

Conclusion:
The demand peak/rush does not drift from the schedule over time. But the latter is structurally blind to the spatial variance of demand peaks.

What Salvador can actually do

Actionable recommendation
Shift the **first morning reinforcement runs ∼20–40 min later** at commuter / hub stops, so buses arrive *with* demand, not ahead of it (i.e. aim for $\kappa(t) \approx \rho(t)$).

- Use **per-stop adaptive thresholds**, not one global window.
- **Stratify by day-type** (weekday / weekend / holiday): commuter stops lose 60–80% of peak volume at weekends.
- The schedule is the one lever the operator directly controls, so schedule re-alignment is the most direct path to impact.
- Use a larger network across an entire year to map different environmental regimes (e.g., seasons).

Limitations & conclusion

Limitations

- 3-month, 10-stop subset, no carnival; no slow seasonal drift to find.
- Rain is a weak covariate (Mar–May is Salvador's wet season, $\sim$74% rainy days).
- Destination stops (e.g. UFBA) carry the morning signal in *alighting*, not boarding.

Conclusion

- Static GTFS windows fail to track structural, stop-specific demand.
- An online CapyMOA pipeline locates the real peaks and quantifies the supply–demand gap, turning raw boardings into a concrete schedule fix.

Thank you!

Questions?

Concept Drifters: Aryan, Eugene, Hans, Necmi, Saurabh, Van